COVID-19 Global Impact Analysis

Executive Summary | 31 December 2019 – 19 September 2020

Objective

This project examines reported COVID-19 cases and deaths across India, the United States, the United Kingdom, Brazil, and Japan. The analysis uses daily data, seven-day rolling averages, population-adjusted case rates, and reported case fatality ratios.

Key Findings

1. The United States recorded the highest cumulative totals in this comparison, with approximately 6.72 million cases and 198,589 deaths by 19 September 2020.

2. India had the highest peak seven-day average of new cases, at approximately 93,199 cases per day.

3. The United States had the highest peak seven-day average of new deaths, at approximately 2,715 deaths per day.

4. Brazil recorded approximately 21,148 cases per million people, slightly above the United States at 20,316 per million.

5. The United Kingdom had the highest reported case fatality ratio, around 10.81%. Differences in testing, reporting, and timing limit direct comparisons.

Methodology

The dataset was filtered to five countries, dates were converted to datetime format, and duplicate country-date records were checked. Seven-day rolling averages were calculated using seven daily observations. Missing values were retained where information was unavailable. The case fatality ratio was calculated as total reported deaths divided by total reported cases, multiplied by 100.

Conclusion and Limitations

The selected countries experienced different reported case and death patterns during the period studied. Total counts and population-adjusted rates provide complementary perspectives. Findings are limited to the available data through 19 September 2020 and may be affected by differences in testing and reporting. The dataset does not contain vaccination data, so no conclusions about vaccination effects are made.

Tools Used: Python, Pandas, Matplotlib, Google Colab